2024, Mar. 2024 Release LOC:Acute Pediatric

Croup

Utilization Benchmarks

Length of Stay: **2.1 d**
Length of Stay Type: **InterQual**
Condition/Procedure: **CROUP**
% Paid as Observation: **63%**

Percent Paid as Observation indicates the final payment status as a percentage for patients hospitalized with the condition. It may not correspond to the status assigned at admission. Note that patients initially placed in Observation status who are converted to inpatient status (common) are included in the inpatient category, and that patients initially admitted to inpatient status who are converted to observation status (uncommon) are included in the Observation category.

Overview

Introduction:
Croup, or laryngotracheobronchitis, is a viral respiratory illness of infants and children and is often contracted in the autumn or winter months. While most cases of croup are mild and can be managed in the outpatient setting, croup can cause severe respiratory obstruction requiring hospitalization.

Evaluation or Treatment:
Diagnosis of croup is based on clinical presentation and is often preceded by symptoms of a mild upper respiratory tract infection. Symptoms of croup include:

- Barking cough
- Inspiratory stridor
- Hoarseness
- Respiratory distress

Croup-like symptoms may be present in children with other conditions such as epiglottitis, bacterial tracheitis, a foreign body of the airway, or a congenital abnormality of the respiratory tract. A high index of suspicion must be maintained for alternate diagnoses, particularly in children with any of the following: toxic appearance, drooling or difficulty swallowing, stridor or respiratory distress (which responds poorly to racemic epinephrine), wheezing or decreased air entry, and rapid deterioration. Children presenting with any of the aforementioned condition may require additional interventions such as early admission to intensive care, intubation, bronchoscopy, and/or IV antibiotics to treat the other disease processes that may be involved.

Corticosteroids are the mainstay of treatment for croup. Current guidelines recommend dexamethasone. Supportive treatment of croup includes racemic epinephrine and oxygen if hypoxia is present. Complications of croup are rare, but include bacterial superinfection, pulmonary edema, pneumothorax, lymphadenitis, and otitis media (Gates et al., Cochrane Database Syst Rev 2018; 8: CD001955; Moraa et al., Cochrane Database Syst Rev 2018; 10: CD006822; Smith et al., Am Fam Physician 2018; 97(9): 575-580; Ortiz-Alvarez, Paediatrics & Child Health 2017, 22: 166-9).

InterQual® criteria are derived from the systematic, continuous review and critical appraisal of the most current evidence-based literature and include input from our independent panel of clinical experts. To generate the most appropriate recommendations, a comprehensive literature review of the clinical evidence was conducted. Sources searched included PubMed, Agency for Healthcare Research and Quality (AHRQ) Comparative Effectiveness Reviews, the National Institute for Health and Care Excellence (NICE), the Centers for Medicare and Medicaid Services (CMS), The Cochrane Library, and The Joint Commission. Other medical literature databases, medical content providers, data sources, regulatory body websites, and specialty society resources may also have been utilized. Relevant studies were assessed for risk of bias following principles described in the *Cochrane Handbook*. The resulting evidence was assessed for consistency, directness, precision, effect size, and publication bias. Observational trials were also evaluated for the presence of a dose-response gradient and the likely effect of plausible confounders. The content is based on a variety of references which are cited at specific criteria points throughout the subset.

(Excludes PO medications unless noted)

Select Day, One:

- **Episode Day 1, One:**
  (Symptom or finding within 24h)
  - **OBSERVATION, Both:** [1]
    - Finding, ≥ **One:**
      - High risk, ≥ **One:** [2]
        - Age < 6 mos [3]
        - Barrier to discharge or arrangements for outpatient services ≤ 24h, ≥ **One:** [4]
          - Limited access to prompt medical care
          - Parent or caregiver inability to provide care
          - Outpatient services unavailable
        - Congenital heart disease [5]
        - History of prematurity < 32 wks [6]
        - History of intubation [7]
        - Neuromuscular disorder [8]
        - Recurrent croup ≤ 3 yrs [9]
        - Upper airway structural abnormality [10]
      - Partial response to initial treatment, **One:** [11, 12, 13]
        - Recurrent stridor at rest after 1 dose of nebulized epinephrine [14]
        - Stridor at rest resolved after 1 dose of nebulized epinephrine and, ≥ **One:**
          - Inadequate oral intake [15]
          - Respiratory rate, **One:** [16]
            - ≥ 50 - 59/min, sustained (age < 3 mos) [17]
            - ≥ 40 - 49/min, sustained (age 3 to < 18 mos) [17]
            - ≥ 30 - 39/min, sustained (age 18 mos to < 4 yrs) [17]
            - ≥ 20-29/min, sustained (age > 4 yrs) [17]
          - Retractions [18]
    - Intervention, ≥ **One:**
      - Corticosteroid (includes inhaled or PO) [19, 20]
      - Nebulized epinephrine at least 1 additional dose [21]
      - IV fluid [22, 23]
      - Oximetry at least every 4h
  - **ACUTE, Both:** [24]
    - Partial response to initial treatment, **One:** [11, 12, 25]
      - Continued symptoms after at least 2 doses of nebulized epinephrine and, ≥ **One:**
        - Recurrent stridor at rest [14]
        - O_2 sat ≤ 92%(0.92) and < baseline [26, 27]
        - Respiratory rate, **One:** [16]
          - ≥ 60/min, sustained (age < 3 mos) [17]
          - ≥ 50/min, sustained (age 3 to < 18 mos) [17]
          - ≥ 40/min, sustained (age 18 mos to < 4 yrs) [17]
          - ≥ 30/min, sustained (age > 4 yrs) [17]
    - Intervention, ≥ **One:**
      - Corticosteroid (includes inhaled or PO) [19, 20]
      - Nebulized epinephrine at least 1 additional dose [21]
      - Oxygen > 21%(0.21) to < 35%(0.35) [28]
  - **INTERMEDIATE, ≥ One:** [29]
    - Chronic ventilation and, **One:** [30]
      - Mechanical ventilation and, ≥ **One:**
        - Increase in baseline FiO_2 > 21%(0.21) to 39%(0.39) [27]
        - Increase in baseline duration of ventilator use [27]
      - NIPPV and, ≥ **One:** [31]

- Increase in baseline FiO_2 > 21%(0.21) to 39%(0.39) [27]
- Increase in baseline duration of NIPPV use [27]

- HFNC and, ≥ **One:** [32]
  - Flow rate, **One:** [33]
    - ≤ 20 L/min (> 10 kg)
    - < 2 L/kg/min (≤ 10 kg)
  - FiO_2 > 21%(0.21) to 39%(0.39) [28]
- Nebulized epinephrine every 3-4h [34]
- Oxygen 35-39%(0.35-0.39) [28]

- **CRITICAL, ≥ One:** [35]
  - Chronic ventilation and, **One:** [30]
    - Mechanical ventilation and, ≥ **One:**
      - Change in baseline setting [27, 36]
      - Increase in baseline FiO_2 ≥ 40%(0.40)
    - NIPPV and, ≥ **One:** [31]
      - Change in baseline setting [27, 37]
      - Increase in baseline FiO_2 ≥ 40%(0.40)
  - HFNC and, ≥ **One:** [32]
    - Flow rate, **One:** [38]
      - > 20 L/min (> 10 kg)
      - ≥ 2 L/kg/min (≤ 10 kg)
    - FiO_2 ≥ 40%(0.40) [28]
  - Mechanical ventilation
  - Nebulized epinephrine every 1-2h or continuous [34]
  - NIPPV [31]
  - Oxygen ≥ 40%(0.40) [28]

- **Episode Day 2, One:**
  - **OBSERVATION, One:**
    - **Responder**, discharge expected today if clinically stable last 12h, **All:** [39]
      - Caregiver able to manage patient at home and follow-up as needed
      - Croup symptoms resolved
      - O_2 sat within acceptable limits [40]
      - Outpatient services arranged or not indicated
      - Tolerating PO [41]
    - **Non-responder**, not clinically stable for discharge and exceeds initial 24h Observation for this condition, **One:**
      - For croup use Episode Day 2 at a higher level of care
      - For a condition other than croup use appropriate subset (Episode Day 1) based on clinical findings
      - Refer for secondary review
  - **ACUTE, One:**
    - **Responder**, discharge expected today if clinically stable last 24h, **All:** [39]
      - Croup symptoms resolved
      - O_2 sat within acceptable limits [40]
      - Tolerating PO [41]
      - No complication or active comorbidity relevant to this episode of care [42]
    - **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**
      - Recurrent stridor and, ≥ **One:** [14]
        - Corticosteroid (includes inhaled or PO) [19, 20]
        - Nebulized epinephrine at least 1 dose [43]
      - Inadequate oral intake and IV fluid [15, 22, 23]
      - Oxygen > 21%(0.21) to < 35%(0.35) [28]
      - Post Critical care ≤ 24h
  - **INTERMEDIATE, ≥ One:**

- Nebulized epinephrine every 3-4h [34]
- ⊖ Chronic ventilation and, ≥ **One:** [30]
  - FiO_2 > 21%(0.21) to 39%(0.39)
  - Respiratory intervention every 3-4h [44]
  - Weaning to baseline ≤ 2d [27]
- ⊖ HFNC and, ≥ **One:** [32, 45]
  - ⊖ Flow rate, **One:** [33]
    - ≤ 20 L/min (> 10 kg)
    - < 2 L/kg/min (≤ 10 kg)
  - FiO_2 > 21%(0.21) to 39%(0.39) [28]
- NIPPV [31]
- Oxygen 35-39%(0.35-0.39) [28]

⊖ **CRITICAL, ≥ One:**
- ⊖ Chronic ventilation and, ≥ **One:** [30]
  - Change in baseline setting [27, 46]
  - FiO_2 ≥ 40%(0.40)
  - Respiratory intervention every 1-2h [44]
- ⊖ HFNC and, ≥ **One:** [32, 45]
  - ⊖ Flow rate, **One:** [38]
    - > 20 L/min (> 10 kg)
    - ≥ 2 L/kg/min (≤ 10 kg)
  - FiO_2 ≥ 40%(0.40) [28]
- Mechanical ventilation
- Nebulized epinephrine every 1-2h or continuous [34]
- ⊖ NIPPV and, ≥ **One:** [31]
  - FiO_2 ≥ 40%(0.40)
  - Respiratory intervention every 1-2h [44]
- Oxygen ≥ 40%(0.40) [28]

⊖ **Episode Day 3, One:**
- ⊖ **ACUTE, One:**
  - ⊖ **Responder**, discharge expected today if clinically stable last 24h, **All:** [39]
    - Croup symptoms resolved
    - O_2 sat within acceptable limits [40]
    - Tolerating PO [41]
    - No complication or active comorbidity relevant to this episode of care [42]
  - ⊖ **Non-responder**, not clinically stable for discharge and exceeds episode days at the Acute level of care for this condition, **One:**
    - **Use Extended Stay subset**
    - Refer for secondary review
- ⊖ **INTERMEDIATE, One:**
  - ⊖ **Non-responder**, Not clinically stable for discharge and exceeds episode days at the Intermediate level of care for this condition, **One:**
    - **Use Extended Stay subset**
    - Refer for secondary review
- ⊖ **CRITICAL, One:**
  - ⊖ **Non-responder**, not clinically stable for discharge and exceeds episode days at the Critical level of care for this condition, **One:**
    - **Use Extended Stay subset**
    - Refer for secondary review

Discharge, Level of Care, One: [47]
- ⊖ **HOME, Both:**
  - ⊖ Level of care appropriateness, **Both:**

- Home environment safe and accessible [48]
- Patient or caregiver demonstrates ability to manage care
- Croup discharge planning, **All:**
  - Education on the importance of recognizing signs and symptoms of worsening airway obstruction and when to seek medical care
  - Provide comprehensive written discharge and teaching instructions [49]
  - Medication reconciliation [50]
  - Follow-up care planned [51]
  - Identify and address transportation needs
- **HOME CARE, Both:**
  - Level of care appropriateness, **All:**
    - Home environment safe and accessible [48]
    - Patient or caregiver willing and able to learn care
    - Outpatient management contraindicated or unavailable [52, 53]
    - Skilled services, ≥ **One:** [54]
      - Skilled nursing [55]
      - Skilled therapy, PT, OT, or ST
      - Behavioral health [56]
  - Croup discharge planning, **All:**
    - Education on the importance of recognizing signs and symptoms of worsening airway obstruction and when to seek medical care
    - Medication reconciliation [50]
    - Provide comprehensive written discharge and teaching instructions [49]
    - Follow-up care planned [51]
    - Identify and address transportation needs
- **BEHAVIORAL HEALTH, Both:**
  - Level of care appropriateness, **One:**
    - Support systems available [57]
  - Skilled treatment, ≥ **One:**
    - Clinical assessment [58]
    - Individual, group, or family counseling
    - Psychiatric, substance, or medication evaluation [59]
- **SKILLED MEDICAL OR THERAPY, Both:**
  - Level of care appropriateness, **All:**
    - Medical practitioner, NP, PA assessment or oversight ≥ 1x/wk
    - Treatment precluded in a lower level
    - Services, ≥ **One:**
      - Medical and skilled nursing interventions or assessment 1-2x/24h [55]
      - Therapy, **All:**
        - Goal directed therapy with rehabilitation potential [60, 61]
        - Functional impairments requiring at least supervision [62]
        - Able to tolerate 1h/d of skilled therapy ≥ 5d/wk
  - Complete prior to facility transfer, **All:**
    - Medication reconciliation [50]
    - Obtain and complete forms for facility
    - Obtain discharge summary and transmit to facility and medical practitioner
    - Arrange transportation
- **SUBACUTE MEDICAL OR THERAPY, Both:**
  - Level of care appropriateness, **All:**
    - Medical practitioner, NP, PA assessment or oversight ≥ 2x/wk
    - Treatment precluded in a lower level
    - Services, ≥ **One:**
      - Medical and skilled nursing interventions or assessment 4h/24h [55]

- Therapy, **All:**
  - Goal directed therapy with rehabilitation potential (60, 61)
  - ≥ 2 functional impairments requiring at least minimum assistance (62)
  - Able to tolerate 2-3h/d of skilled therapy ≥ 5d/wk and ≥ 1 therapy discipline
- Complete prior to facility transfer, **All:**
  - Medication reconciliation (50)
  - Obtain and complete forms for facility
  - Obtain discharge summary and transmit to facility and medical practitioner
  - Arrange transportation

- **SUBACUTE REHAB, Both:**
  - Level of care appropriateness, **All:**
    - Medical practitioner, NP, PA assessment or oversight ≥ 3x/wk
    - Treatment precluded in a lower level
    - Services, **All:**
      - Goal directed therapy with rehabilitation potential (60, 61)
      - ≥ 2 functional impairments requiring at least minimum assistance (62)
      - Able to tolerate 2-3h/d of skilled therapy ≥ 5d/wk and ≥ 2 therapy disciplines
  - Complete prior to facility transfer, **All:**
    - Medication reconciliation (50)
    - Obtain and complete forms for facility
    - Obtain discharge summary and transmit to facility and medical practitioner
    - Arrange transportation
- **ACUTE REHAB, Both:**
  - Level of care appropriateness, **Both:**
    - Medical practitioner, NP, PA assessment or oversight ≥ 3x/wk
    - Services, **All:**
      - Goal directed therapy with rehabilitation potential (60, 61)
      - ≥ 2 functional impairments requiring at least minimum assistance (62)
      - Able to tolerate ≥ 15h/wk of skilled therapy and ≥ 2 therapy disciplines (63)
  - Complete prior to facility transfer, **All:**
    - Medication reconciliation (50)
    - Obtain and complete forms for facility
    - Obtain discharge summary and transmit to facility and medical practitioner
    - Arrange transportation
- **LONG TERM ACUTE CARE, Both:**
  - Level of care appropriateness, **Both:**
    - Treatment precluded in a lower level
    - In lieu of acute or continued hospitalization or failed lower level of care, ≥ **One:**
      - Primary condition or illness and treatment of active comorbid condition(s) (64)
      - Ventilator dependent ≥ 6h/d and weaning planned (65, 66)
      - Acute and comorbid conditions requiring prolonged hospitalization (67)
  - Complete prior to facility transfer, **All:**
    - Medication reconciliation (50)
    - Obtain and complete forms for facility
    - Obtain discharge summary and transmit to facility and medical practitioner
    - Arrange transportation

Notes:

1:

Hemodynamically stable patients who require at least 6 hours, and for certain conditions, up to 48 hours, of treatment or assessment pending a decision regarding the need for additional care. Observation level of care services may be provided in designated observation units or on a hospital floor.
Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

2:

Instruction: Patients with the following criteria could qualify for Observation level of care due to their high risk for developing complications and may require further monitoring.

3:

The peak incidence of croup occurs in children between the ages of 6 months and 3 years but can be seen in children up to 6 years of age. Croup in young infants less than 6 months of age is considered atypical and raises the possibility of different etiology other than viral infection causing an upper airway obstruction. Further monitoring and workup maybe indicated (Hanna et al., Int J Pediatr Otorhinolaryngol 2019, 127: 109686; Petrocheilou et al., Pediatr Pulmonol 2014, 49: 421-9).

4:

There is limited evidence to recommend absolute barriers to discharge for pediatric patients. Literature does support that children with limited access to prompt medical care, especially children who do not have a primary care provider, are at increased risk of readmission and noncompliance with prescription regimens. Children who come from limited-income households or have a parent or caregiver who is physically or mentally unable to care for them are also at higher risk of not getting the necessary follow-up care if discharged from the hospital without social service involvement (Johns Hopkins University and Quality, Improving the Emergency Department Discharge Process: Environmental Scan Report. 2014).

5:

There are limited data in the literature linking congenital heart disease to high risk of severe croup; however, it has been noted as one of the comorbidities that could increase the need for hospitalization in patients with croup (Pound et al., Hospital Pediatrics 2020, 10: 1068-77). Congenital heart disease can lead to compromised airway due to compression from malformed or enlarged cardiovascular structures. Also, congenital heart disease, especially unrepaired defects, can lead to chronic pulmonary problems such as pulmonary edema and pulmonary hypertension which places these patients at high risk for complication (Healy et al., Paediatr Respir Rev 2012, 13: 10-5).

6:

Prematurity has been associated with increased hospitalization in children with croup. This may be related to comorbid conditions like chronic lung disease or a history of prolonged intubation. Prolonged intubation can cause upper airway abnormalities, such as subglottic stenosis, and lead to more severe croup symptoms (Pound et al., Hospital Pediatrics 2020, 10: 1068-77).

7:

History of intubation is a risk factor associated with airway abnormalities resulting in increased incidence for croup hospitalization (Quraishi and Lee, Pediatr Clin North Am 2022, 69: 319-28; Pound et al., Hospital Pediatrics 2020, 10: 1068-77).

8:

Children with a neuromuscular disorder often have weak and poorly coordinated respiratory muscles, stiff chest wall with reduced chest mobility, inability to effectively cough to clear secretions, and recurrent infections leading to chronic inflammation of airway. This puts them at high risk of complication with any respiratory infection

(DellaBadia and Tauber, Curr Probl Pediatr Adolesc Health Care 2021, 51: 101072).

9:

Recurrent croup, defined as ≥ 2 episodes per year or > 3 episodes before the age of 3 years old, has been identified as a risk factor for airway abnormality (Quraishi and Lee, Pediatr Clin North Am 2022, 69: 319-28).

10:

Upper airway abnormalities are risk factors associated with severe croup, prolonged symptoms, and/or recurrent croup. Such airway abnormalities can include, but not limited to, subglottic stenosis, tracheomalacia, laryngomalacia, and laryngeal hemangioma (Quraishi and Lee, Pediatr Clin North Am 2022, 69: 319-28; Pound et al., Hospital Pediatrics 2020, 10: 1068-77; Hanna et al., Int J Pediatr Otorhinolaryngol 2019, 127: 109686).

11:

The mainstay of treatment for croup is corticosteroids, which can be given orally, intravenously, intramuscularly, or inhaled, and can improve symptoms of croup in as early as 2 hours after the administration. Nebulized epinephrine also improves symptoms of croup with quicker onset of action; however, its effect is short lasting. It is used as adjunctive therapy in moderate to severe croup until corticosteroid takes full effect. Children with mild symptoms may not need nebulized epinephrine prior to discharge from the emergency department (Gates et al., Cochrane Database Syst Rev 2018; 8: CD001955; Smith et al., Am Fam Physician 2018; 97(9): 575-580; Ortiz-Alvarez, Paediatrics & Child Health 2017, 22: 166-9).

12:

Mild croup is characterized by mild or no stridor at rest, an occasional barky cough, and mild chest wall retractions. Moderate croup is characterized by the presence of stridor at rest along with barky cough and chest wall retractions. In severe croup, patients can have prominent stridor and chest wall retractions accompanied by agitation or lethargy (Smith et al., Am Fam Physician 2018; 97(9): 575-580; Ortiz-Alvarez, Paediatrics & Child Health 2017, 22: 166-9).

13:

Initial treatment(s) for croup may include:

- Corticosteroids (a systemic dose given within the past 24 hours in any outpatient setting)
- At least 1 dose of nebulized epinephrine administered in any outpatient setting prior to decision to admit

Outpatient management is inclusive of treatment administered in a medical practitioner's office, outpatient clinic (e.g., Urgent Care center), or emergency department.

14:

Stridor is one of the main symptoms of croup and results from air moving through a narrowed upper airway. In croup, the edema from the inflammation of the airway mucosa due to viral respiratory infection leads to the narrowing and obstruction of the upper airway (Smith et al., Am Fam Physician 2018; 97(9): 575-580; Ortiz-Alvarez, Paediatrics & Child Health 2017, 22: 166-9).

15:

Inadequate oral intake refers to the inability to maintain hydration according to a patient's documented fluid goal or maintenance fluid requirement. The inability to maintain hydration may occur when excessive output (e.g., vomiting, diarrhea, or insensible losses) leads to a net fluid deficit.

16:

Vital signs are considered sustained findings when they are abnormal for 2 or more readings greater than 15 minutes apart or are lasting at least 15 minutes. This excludes an isolated reading, a transient abnormal measurement, or a finding that requires urgent treatment (e.g., severe hypoxia, ventricular arrhythmia, or hypotension). Application of criteria for a vital sign finding should be based on confirmatory measurements repeated at regular intervals. Events that are infrequent or vital sign abnormalities that have resolved with outpatient treatment are not considered to be sustained (e.g., the resolution of hypertension following anti-

hypertensive therapy or the resolution of tachycardia following rehydration).
Note: The definition of sustained reflects the opinion of InterQual's® expert clinical consultants. The criteria are based upon current best practice and are the product of an iterative process involving multiple clinicians with diverse expertise in varied practice and geographic settings.

17:

Vital signs are considered sustained findings when they are abnormal for 2 or more readings greater than 15 minutes apart or are lasting at least 15 minutes. This excludes an isolated reading, a transient abnormal measurement, or a finding that requires urgent treatment (e.g., severe hypoxemia, ventricular arrhythmia, hypotension). Application of criteria for a vital sign finding should be based on confirmatory measurements repeated at regular intervals. Events that are infrequent or vital sign abnormalities that have resolved with outpatient treatment are not considered to be sustained (e.g., resolution of hypertension following anti-hypertensive therapy, resolution of tachycardia following rehydration).
Note: The definition of sustained reflects the opinion of InterQual's® external peer reviewers. This criteria point is based upon current best practice and is the product of an iterative process involving multiple clinicians with diverse expertise in varied geographic settings.

18:

The presence and severity of retraction is often used to determine the severity of croup symptoms in children. The Westley Croup score, one of the most commonly used tools in literature, evaluates croup severity based on five categories: the level of consciousness, retractions, cyanosis, stridor, and air entry (Smith et al., Am Fam Physician 2018; 97(9): 575-580). Expert consensus guideline from Toward Optimized Practice introduced a different way to describe croup severity based on the presence of suprasternal and/or intercostal retractions, barky cough, stridor at rest, cyanosis, and mental status changes (distress, agitation, or lethargy) (Ortiz-Alvarez, Paediatrics & Child Health 2017, 22: 166-9).

19:

A Cochrane Review found that corticosteroids are effective in improving the symptoms of croup as early as two hours after treatment. Additionally, corticosteroid administration is associated with fewer return visits to the Emergency Department (ED) and a shorter length of hospital stay (Gates et al., JAMA Pediatr 2019; 173(6): 595-596; Gates et al., Cochrane Database Syst Rev 2018; 8: CD001955; Ortiz-Alvarez, Paediatrics & Child Health 2017, 22: 166-9).

20:

Instruction: Inhaled corticosteroid can be applied to this criterion if inhaled corticosteroid is a new medication for the patient (i.e., patient is not already on inhaled corticosteroid at home).
Corticosteroids can be administered orally, intravenously, intramuscularly, or inhaled. Dexamethasone is the most studied corticosteroid for croup. It is usually given as a single dose and can be given orally. However, other routes may be necessary if a patient with croup is vomiting or unsafe to take oral medication. When other forms of corticosteroid (such as prednisolone) are used, a longer course of treatment may be required (Parker and Cooper, Pediatrics 2019, 144:; Gates et al., Cochrane Database Syst Rev 2018; 8: CD001955; Ortiz-Alvarez, Paediatrics & Child Health 2017, 22: 166-9).

21:

Nebulized dose here refers to additional nebulized epinephrine given after the decision to admit.

22:

To allow for the advancement of feedings as intravenous (IV) fluids are weaned, the IV fluid rates are ≥ 50%(0.50) of the maintenance fluid requirement. These rates can be calculated using the following (National Clinical Guideline Centre for Acute and Chronic Conditions, IV Fluids in Children: Intravenous Fluid Therapy in Children and Young People in Hospital. 2015. Reaffirmed 2020):
Weight ≤ 10 *kg* = ≥ 50 *mL/kg*/24h

- Hourly rate = 2.08 (*mL/h*) x kg

Weight > 10-25 *kg* = ≥ 40 *mL/kg*/24h

- Hourly rate = 1.67 (*mL/h*) x kg

Weight > 25-60 *kg* = ≥ 30 *mL/kg*/24h
- Hourly rate = 1.25 (*mL/h*) x kg

Example: An 11 *kg* child is required to receive a minimum of 40 *mL/kg*/24h to meet this criteria point. Multiply 1.67 (*mL*/h) x 11 (child's weight in *kg*) = 18.37 *mL*/h. This is the minimum fluid replacement rate for this child.

23:

Instruction: This line of criteria is intended for the patient receiving condition-directed maintenance IV fluid and excludes the patient who has IV fluid ordered to keep vein open (KVO).

24:

Hemodynamically stable patients who require treatment, assessment, or intervention every 4 to 8 hours.
Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

25:

Initial treatments include:
- Corticosteroids (a systemic dose given within the past 24 hours in any outpatient setting)
- At least 2 doses of nebulized epinephrine administered in any outpatient setting prior to decision to admit

Outpatient management is inclusive of treatment administered in a medical practitioner's office, outpatient clinic (e.g., Urgent Care center), or emergency department.

26:

Supplemental oxygen treatment is recommended to maintain an oxygen saturation greater than 92% for the pediatric patient with croup (Scarfone and Seiden, Rosen's emergency medicine : concepts and clinical practice, Ninth edition. ed. 2018)

27:

Baseline refers to either the patient's normal baseline or a newly established baseline. In the absence of documentation, a patient's baseline status may be presumed to be normal.

28:

Oxygen therapy is the administration of oxygen at concentrations greater than ambient air [room air: 21%(0.21) FIO_2] to correct hypoxia. For pediatric patients, the selection of an oxygen delivery system needs to take into consideration the patient's size, needs, and therapeutic goals. The oxygen concentration or percentage (FIO_2) delivered varies with the manufacturer's design, delivery device, patient's minute ventilation (dependent on the respiratory rate and tidal volume), and oxygen flow rate. In general, patients with higher minute ventilation (common in most sick patients and those with respiratory disease) require considerably higher flow rates to correct hypoxia. The following are examples of oxygen delivery systems commonly used for pediatric patients with the percentage range of FIO_2 optimally delivered:
Low-flow systems
- Nasal cannula / Nasopharyngeal catheters 24-35%(*0.24-0.35*) FIO_2

Reservoir systems (For neonates, infants, and young children, the maximum FIO_2 achievable is not well documented)
- Simple oxygen masks 35-50%(*0.35-0.50*) FIO_2
- Partial-rebreathing masks 40-60%(*0.40-0.60*) FIO_2
- Non-rebreathing masks > 60%(*0.60*) FIO_2

High-flow systems

- Air-entrainment masks / nebulizers 24-40%(*0.24-0.40*) FIO_2
- Heated, humidified, high flow oxygen concentrating nasal cannula 21-100%(*0.21-1.0*) FIO_2

Enclosure systems

- Oxygen hoods / tents 21-100%(*0.21-1.0*) FIO_2

For children less than 5 years old, the oxygen regimen may need to be adjusted to a lesser concentration and flow rate, while still requiring acute level of care due to the patient's size, metabolic needs, and vital sign instability.

29:

Hemodynamically stable patients who require treatment, assessment, or intervention every 2 to 4 hours.
Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

30:

Chronic ventilation refers to a patient who is managed at home on chronic mechanical ventilation via a tracheostomy tube or noninvasive positive pressure ventilation via a mask. This includes continuous positive airway pressure (CPAP) or bilevel positive airway pressure (BiPAP).

31:

Noninvasive positive pressure ventilation (NIPPV), also known as NIV, provides respiratory support by application of a tightly fitting facial mask, nasal mask, or helmet rather than an endotracheal tube or tracheostomy. In some cases, the use of NIPPV can avoid the need for endotracheal intubation and decreases the risk of barotrauma, lung injury, and/or infection. NIPPV is commonly delivered by a bilevel positive airway pressure ventilator (BiPAP), a continuous positive airway pressure device (CPAP), or a mechanical ventilator. Supplemental oxygen can be delivered at concentrations approximating 100%(1.0).
The decision to provide respiratory support via NIPPV, and the modality to provide NIPPV, is based upon the patient's specific clinical findings (e.g., medical condition leading to respiratory failure, underlying comorbidities, clinical progression, improvement). Examples of NIPPV devices are volumetric (i.e., deliver a defined volume), barometric (i.e., deliver a defined pressure), and combined (i.e., deliver defined volumes and pressure). The terminology for NIPPV delivery systems may differ between equipment manufacturers and provider organizations. InterQual® criteria do not differentiate between the different NIPPV modalities.
Whether or not high-flow nasal cannula (HFNC) should be classified as a component of NIPPV is unclear, and there is conflicting evidence in the medical literature. Where HFNC is an appropriate modality, InterQual® defines this in a specific criteria point. As such, NIPPV does not include HFNC (Hackett, A., PulmCCM 2018; Nardi et al., F1000Res 2017, 6: 290; Osadnik et al., Cochrane Database Syst Rev 2017, 7: CD004104; Allison and Winters, Emerg Med Clin North Am 2016, 34: 51-62; Gregoretti et al., Crit Care Clin 2015, 31: 435-57).

32:

High flow nasal cannula (HFNC) delivers a fraction of inspired oxygen (FiO_2) that is heated (up to 37 degrees Celsius) with 100%(1.0) relative humidity at flow rates set independent from the FiO_2. The heated and humidified gas has been shown to improve comfort, aid in the clearance of secretions, and reduce bronchospasms. When flow rates are set above the patient's inspiratory demand, the upper airway is washed out of carbon dioxide, nasal resistance is decreased, and there is a reduction in dead space, leading to improved work of breathing. In addition, HFNC reduces the amount of ambient air a patient can breathe in, resulting in less oxygen dilution. Bronchiolitis is the main indication for HFNC in pediatric patients beyond the neonatal period. Although the benefits of HFNC in other conditions such as pneumonia, asthma, and post-extubation have been reported, more well-designed studies are needed to guide the use of HFNC in these conditions (Kwon, Clin Exp Pediatr 2020, 63: 3-7).

33:

High flow nasal cannula (HFNC) consists of three parameters: temperature, fraction of inspired oxygen (FiO_2), and flow rate. Each parameter is independent of the other. The flow rate is how fast the heated and humidified air/oxygen is delivered per minute and is documented as liters per minute (L/min). Standard HFNC flow rates in

pediatrics have yet to be established (Kwon, Clin Exp Pediatr 2020, 63: 3-7). Although most studies on HFNC in pediatrics focus on children with bronchiolitis, they can provide some guidance on appropriate flow rates outside of the intensive care unit (ICU). Initial flow rates in clinical trials have found that 1-2 L/kg/min is effective in patients with bronchiolitis (Yurtseven et al., Pediatr Pulmonol 2019, 54: 894-900). However, a recent survey revealed that only 19%(0.19) of responders use a weight-based flow rate method (Miller et al., Respir Care 2018, 63: 894-9). This method may not be optimal in older children as it can result in very high flow rates that may not be tolerated. Some studies combat this potential problem by setting maximum flow rates. Flow rates up to 20 L/min are well tolerated on non-ICU wards (Willer et al., Hosp Pediatr 2021, 11: 891-5; Kepreotes et al., Lancet 2017, 389: 930-9). Therefore, a flow rate of 20 L/min was chosen as the maximum for the Intermediate level of care.
Instruction: To calculate the L/kg/min, the flow rate is divided by the patient's weight.
Example:
HFNC flow rate: 12 L/min
Patient weight: 8 kg
12/8 = 1.5
This patient is on a flow rate of 1.5 L/kg/min

34:

Nebulized epinephrine usually results in improvement of respiratory distress within 30 minutes or sooner. Its effect, however, is transient, lasting about 2 hours, and it is mostly used to manage symptoms until the corticosteroid effect takes place (Udoh et al., Wmj 2022, 121: 26-9; Ortiz-Alvarez, Paediatrics & Child Health 2017, 22: 166-9; Rudinsky et al., The Journal of Emergency Medicine 2015, 49: 408-14).

35:

Hemodynamically unstable patients (or those with the potential to become unstable) who require treatment, assessment, or intervention every 1 to 2 hours.
Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

36:

A change in 1 or more of the following settings from baseline would meet at the Critical level of care:

- Inspiratory time (iTime)
- Positive end-expiratory pressure (PEEP)
- Pressure support (PS)
- Respiratory rate
- Tidal volume (TV)

37:

A change in 1 or more of the following settings from baseline would meet at the Critical level of care:

- Inspiratory positive airway pressure (IPAP)
- Expiratory positive airway pressure (EPAP)
- Positive end expiratory pressure (PEEP)

38:

High flow nasal cannula (HFNC) consists of the following three parameters: temperature, fraction of inspired oxygen (FiO_2), and flow rate. Each parameter is independent of each other. The flow rate is how fast the heated and humidified air/oxygen is delivered per minute and is documented as liters per minute (L/min). The flow rate is often based on the patient's weight in smaller children (Kwon, Clin Exp Pediatr 2020, 63: 3-7). To calculate the liters per kilogram per minute (L/kg/min), the flow rate is divided by the patient's weight.
Example:
HFNC flow rate: 12 L/min
Patient weight: 8 kg
12/8 = 1.5
This patient is on a flow rate of 1.5 L/kg/min

39:

Selection of this criteria point indicates that the patient is responding to treatment and is clinically stable for transfer or discharge. To determine the most appropriate post-acute level of care, see discharge criteria.

40:

When a criteria point states "within acceptable limits," it refers to either the patient's normal baseline, a newly established baseline, or parameters that the medical practitioner determines are acceptable.

41:

Oral intake should be greater than or equal to the patient's maintenance fluid requirements or within parameters which the medical practitioner determines are acceptable. Maintenance fluid requirements should be increased in the presence of dehydration or insensible loss. Maintenance fluid requirements (*mL/kg*/d) can be calculated based on current body weight (*kg*):

- 100 *mL/kg*/d for first 10 *kg*
- 50 *mL/kg*/d for second 10 *kg*
- 20 *mL/kg*/d for each additional *kg* in weight

42:

Relevant complications or comorbidities are conditions that actively require management during an inpatient stay.

43:

The mainstay of treatment for croup is corticosteroids, which can be given orally, intravenously, intramuscularly, or inhaled, and can improve symptoms of croup in as early as 2 hours after the administration. Nebulized epinephrine also improves symptoms of croup with quicker onset of action, however its effect is short lasting. It is used as adjunctive therapy in moderate to severe croup until corticosteroid takes full effect (Gates et al., Cochrane Database Syst Rev 2018; 8: CD001955; Smith et al., Am Fam Physician 2018; 97(9): 575-580; Ortiz-Alvarez, Paediatrics & Child Health 2017, 22: 166-9).

44:

Respiratory interventions include ventilator/noninvasive positive pressure ventilation (NIPPV) setting changes, chest physiotherapy, suctioning, and nebulizer or metered-dose inhaler (MDI) treatments.

45:

Currently, there is no universally accepted method for weaning pediatric patients off high-flow nasal cannula (HFNC), and weaning protocols vary based on institutional policies. However, one commonly used method is to gradually decrease the fraction of inspired oxygen (FiO_2) to a specific level, such as 40%(0.40), then gradually reduce the flow rate before transitioning to a low-flow oxygen delivery system (e.g., nasal cannula). Less common methods include weaning the FiO_2 to a predetermined value, then transitioning to a low-flow oxygen system without reducing the flow rate, or gradually reducing the flow rate while keeping the FiO_2 constant (Kawaguchi et al., Pediatr Crit Care Med 2020, 21: e228-e35). There is no standard flow rate at which HFNC should be discontinued, but based on the literature, it is suggested to discontinue HFNC when the flow rate reaches 2 L/min in infants, and 4 L/min in children older than 12 months (Udurgucu et al., Pediatr Allergy Immunol Pulmonol 2022, 35: 79-85; Peterson et al., Respir Care 2021, 66: 591-9).
Instruction: When weaning a patient off high-flow nasal cannula (HFNC), it is expected that the current level of care be maintained or a lower level of care from the previous level be utilized to conduct the review if HFNC criteria are used. A typical weaning strategy usually does not involve simultaneously decreasing the flow rate while increasing the FiO_2. This scenario may indicate that a patient is being kept on the HFNC machine but may be able to tolerate settings that could be administered by nasal cannula. For instance, a patient who was previously on HFNC 10 L/min, 30%(0.30) FiO_2 might be weaned to 3 L/min, but FiO_2 is increased to 100%(1.0). In this case, the patient should be assessed based on the flow rate rather than the FiO_2.

46:

A change in 1 or more of the following settings from baseline would meet at the Critical level of care:

- Expiratory positive airway pressure (EPAP)

- Inspiratory positive airway pressure (IPAP)
- Inspiratory time (iTime)
- Positive end expiratory pressure (PEEP)
- Pressure support (PS)
- Respiratory rate (RR)
- Tidal volume (TV)

47:

The discharge screen is a resource tool and not criteria. Referring to the discharge screen at the initiation of discharge planning is recommended.

48:

The home environment and safety assessment will normally include an evaluation of the patient's pre- and post-hospitalization functional level, physical layout of the home (e.g., entrances and exits, stairs, access to the community), identification of unsafe conditions (e.g., scatter rugs, missing handrails, oxygen use and smoking, lack of fire safety devices, inadequate lighting, heating, and cooling), factors that may trigger symptoms (e.g., secondhand smoke, poor food choices, dysfunctional family dynamics), sanitation hazards (e.g., lack of electricity, running water, refrigeration, inadequate toilet facilities, presence of insects or rodents), and the need for adaptive equipment (e.g., walker, handrails for the tub or shower, elevated toilet seat).

49:

Ensuring the patient and/or caregiver understands all aspects of the condition and can assume responsibility for self-care is crucial in preventing readmission. Assessing a patient and/or caregiver's level of understanding after education has been provided can be difficult. The teach-back method is an effective way of providing education at the appropriate level and can also be used to assess the learner's comprehension. To use the teach-back method, discuss key points in common terms and avoid using medical jargon or unfamiliar terms. After the education has been delivered, ask the learner to repeat what was learned. Gaps or misinterpretations in the learner's explanation will pinpoint areas where communication may have failed and provide opportunity for clarification.

50:

Medication reconciliation is a formal process or technique used by health care providers and pharmacists to identify the most complete and accurate list of all medications a patient is taking at times of transitions in care (e.g., upon hospital admission, transfer from one unit to another during hospitalization, or discharge from the hospital to home or another facility). The goals of this process are to ensure medication and dosages are appropriate for the patient, resolve discrepancies in drug regimens, and ultimately prevent medication errors and reduce adverse drug events. Medication reconciliation is a Joint Commission National Patient Safety goal. Coordinating information when a patient is transferred to another setting, service, practitioner, or level of care ensures accurate medications are listed. The process is comprised of the following (Hospital: National Patient Safety Goals. 2020):

- Obtain an external list of medications (e.g., medications taken prior to admission)
- Develop a list of current medications and add them to the medical record
- Compare the medications from the external list to the current list
- Clarify inconsistencies/discrepancies (e.g., omissions, duplications, contraindications, unclear information, and changes)
- Develop a list of medications to be prescribed at discharge or transfer
- Communicate the new list to the patient and/or appropriate caregiver(s)
- Ensure that patient and/or caregiver(s) understand the medication information upon discharge

Specific medication issues identified as being problematic include missing medication information from transfer orders, lack of information on medications provided in the acute setting, incomplete medication records, discrepancies between hospital regimen and discharge summary, and missing information pertaining to the patient's tolerance of a medication regimen. Although medication reconciliation is an important aspect in patient safety, there is a lack of consensus and evidence about the best effective methods of implementing this process. Physician-led and electronic medication reconciliation in hospitals are effective strategies to reduce medication discrepancies, however, the impact of these interventions is uncertain due to the low quality of evidence (Choi and Kim, J Clin Pharm Ther 2019, 44: 932-45; Redmond et al., Cochrane Database Syst Rev 2018, 8: CD010791). Trained pharmacy technicians, under the direction of a licensed pharmacist, may be an option for developing and expanding medication reconciliation processes (Irwin et al., Hosp Pharm 2017, 52: 44-53).

51:

Close follow-up of older adults, pediatric, or at-risk patients after discharge can help minimize hospital readmission and total health care costs (Rising et al., Ann Emerg Med 2013, 62: 145-50; Frei-Jones et al., Pediatr Blood Cancer 2009; 52(4): 481-485; Kripalani et al., JAMA 2007; 297(8): 831-841). Studies have shown that for greater than 50% (0.50) of patients re-hospitalized within 30 days of discharge, there was no visit to a physician's office between the time of discharge and re-hospitalization (Hernandez et al., JAMA 2010, 303: 1716-22; Jencks et al., N Engl J Med 2009; 360(14): 1418-1428). Efforts to reduce readmissions should focus on patients known to be at high risk and patients who have frequent visits to the emergency department (ED). Patients at high risk may have multiple ED visits prior to readmission (Rising et al., Ann Emerg Med 2013, 62: 145-50).

52:

Outpatient management contraindicated refers to the following:

- The patient whose medical condition is so fragile that going to the clinic or medical practitioner's office for treatment would put the patient at risk
- The patient with extensive equipment (e.g., ventilators, specialized beds) where travel outside the home would be cumbersome and places the patient at risk
- The treatment goal is to adapt the home environment for increased independence (e.g., an individual who is blind who needs mobility education)

53:

Outpatient management unavailable refers to the following:

- The patient whose treatment (e.g., dressing changes) is so extensive that it takes too long or is too frequent (e.g., twice daily) to be done in an outpatient clinic or medical practitioner's office
- The scheduling of the treatment falls outside the medical practitioner's office or clinic hours
- Travel distance to receive care is excessive (e.g., greater than 1 hour)
- The service is unavailable in the outpatient setting

54:

Skilled nursing services refer to services that must be provided by a licensed nurse qualified to assess and monitor patient condition(s) and provide medical treatment and/or teaching to patients who have skilled nursing needs. Examples of Home Care level of care interventions include skilled assessment, intervention, or education for any of the following:

- New treatment or medication regimen
- Adjustment in the treatment or medication regimen
- Infusion therapy administration, teaching, or access device management
- Management and evaluation of a care plan for patients with an active comorbidity and multiple care needs
- Chronic disease requiring a disease management program (e.g., asthma, heart failure, chronic obstructive pulmonary disease, diabetes)
- End-stage disease, hospice, or palliative care

55:

Skilled nursing interventions refer to services that are provided or supervised by a qualified or licensed professional when the care needs of the patient require assessment, observation, monitoring, and/or education to achieve the medically desired result. Close observation and assessment by nursing may be necessary at this level of care. This includes nursing care such as medication administration (excludes routine), vital signs monitoring, and/or other interventions not listed.

56:

The Home Care criteria are used for a patient who has been admitted or is expected to be admitted for home care. When criteria are met for this level of care and it is not available in your area, we recommend you use the next higher level of care.
Home care can include specialty services provided by a psychiatric nurse following a specialized treatment plan developed by a psychiatrist or advanced practice registered nurse. It is intended for treatment and monitoring of a patient's psychiatric condition that prevents the patient from leaving the house or makes leaving the house

extremely difficult or unsafe (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022). Home care can also be used for a patient with severe and persistent mental illness with a history of nonadherence to medication.

Evaluation and Treatment

Programming may differ based upon legislative and geographical variances and is subject to organizational policy; however, at a minimum, it should include:

- Care coordination with treating physician and other providers
- Clinical assessment every visit
- Education every visit
- Medication assessment, teaching, and management every visit
- Medication reconciliation initiated within first visit
- Psychosocial assessment within first visit
- Substance evaluation within first 2 visits

For those patients who are covered under a Medicare benefit, the Affordable Care Act mandates that a certifying physician or allowed practitioner have a face-to-face visit with the patient within 90 days prior to the start of care, or within 30 days after the start of care. There must be documentation of the patient's condition that supports the need for home care services and the requirement of homebound status (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022).

57:

Support system includes friends, family, social services, health care delivery providers or agencies, sponsors, clergy, and self-help groups.

58:

The initial clinical assessment is performed by a licensed professional and includes a comprehensive review of the patient's presenting diagnosis and a review of body systems. Identification of current and potential medical needs and health problems can be identified by the clinical assessment. The clinical assessment includes:

- History and physical exam (e.g., vital signs, height, weight)
- Pain assessment includes cause, intensity, quality, onset, duration, and effects on quality of life (e.g., activities of daily living (ADLs), instrumental activities of daily living (IADLs), and interpersonal relationships). Pain relievers should also be included
- Nutritional and hydration status
- Functional ability
- Safety and infection control measures
- Prescribed and over-the-counter (OTC) medications, herbal supplements, and home remedies
- Patient's and/or caregiver's understanding of medication use, dosing, side effects, and adherence to the medication or treatment regimen
- Patient's and/or caregiver's understanding of the illness or disease process and potential long-term complications
- Patient's and/or caregiver's coping strategies to deal with the illness or injury and ability to follow the plan of care

59:

For specific level of care recommendations for the treatment of psychiatric and substance use disorders, refer to the InterQual® Adult and Geriatric Psychiatry, InterQual® Child and Adolescent Psychiatry, or InterQual® Substance Use Disorders products.

60:

Goal-directed therapy includes one or more of the following:

- Activities of daily living (ADLs) (e.g., feeding, dressing, bathing, grooming, toileting, functional mobility)
- Bed mobility or transfers (e.g., transfers to chair, toilet, commode, tub, shower, car)
- Cognitive, speech, language, or swallowing retraining
- Home and lifestyle modification (e.g., assessment and training focused on the home, job, school, or work environment)
- Pain management techniques (e.g., assessment and interventions used to relieve pain or decrease its intensity)

- Pulmonary rehabilitation (e.g., energy conservation, speech, breathing re-education, postural drainage, percussion, or vibration)
- Positioning and splinting
- Range of motion and strengthening
- Wheelchair mobility
- Ambulation

61:

Rehabilitation potential refers to the probability that therapy and medical goals are realistic and attainable based on patient's prior level of function, severity of illness or injury, and the extent of impairments.

62:

Functional assistance levels are based upon the patient's function during tasks and activities necessary to return to household mobility or ambulation. The functional assistance level required for each individual task or activity (e.g., mobility, activities of daily living (ADLs)) may vary.
The following terms are commonly used in the post-acute setting:

- Independent - Patient can safely and within a reasonable amount of time perform a task (or developmentally appropriate task) without physical or cognitive assistance or supervision
- Modified Independent - Patient performs an activity with a supportive device, adaptive equipment, and/or prosthetic or orthotic device. Additional time may be required to complete the activity and/or there are safety (risk) considerations
- Supervision - Patient performs an activity with standby or distant supervision or setup. Verbal cueing or coaxing, without physical contact or setup of items and application of orthoses may be required when patient's safety awareness is impaired
- Minimum or limited Assistance - Patient performs at least 75%(0.75) of an activity and requires some physical contact to steady, guide, or move
- Moderate or extensive Assistance - Patient performs at least 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Maximum Assistance - Patient performs 25%(0.25) to 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Total Assistance or dependence - Patient performs less than 25%(0.25) of an activity and may require total assistance for functional mobility or ADLs

63:

The intensity of rehabilitation therapies is a driver for admission to an acute inpatient rehabilitation facility. Documentation should reflect the need for required and ongoing intensive therapies. The determination of whether a patient requires and can tolerate an intensive rehabilitation program of at least 15 hours is not based on any one single piece of documentation. All clinical documentation should be evaluated by the reviewer to determine whether the patient is demonstrating the need for services and has the ability to tolerate them. Patients who do not require this level of intensive therapy should be treated at a lower level of care. Rehabilitation therapies should be reasonable and medically appropriate and should not exceed a patient's level of tolerance. If there are clinical features which preclude the patient's ability to tolerate the intensity threshold of at least 15 hours per week, secondary review would be appropriate (Centers for Medicare & Medicaid Services, Medicare Benefit Policy Manual Chapter 1 - Inpatient Hospital Services Covered Under Part A. 2021; Miller and Forer, Pm r 2021, 13: 535-9; Forrest et al., Medicine (Baltimore) 2019, 98: e17096).

64:

Comorbid conditions include new onset or worsening behavioral symptoms, heart failure, deep vein thrombosis, uncontrolled diabetes, immunocompromised states, malignant or end-stage disease, malnutrition, renal insufficiency or end-stage renal disease, infection, ventilator dependence, noninvasive positive pressure ventilation or respiratory insufficiency, complex wound care, and new functional impairment(s) with limitation.

65:

Ventilation includes invasive ventilation and noninvasive positive pressure ventilation (NIPPV), including continuous positive airway pressure (CPAP) and bilevel positive airway pressure (BiPAP).

66:

Notable causes for prolonged mechanical ventilation include:

- Ventilator-acquired pneumonia
- Renal failure
- Heart failure
- Exacerbation of chronic obstructive pulmonary disease (COPD)
- Chest wall disorder
- Neuromuscular diseases
- Traumatic brain injury
- Sepsis
- Complications following cardiac surgical procedures

Major contributing factors for failure to wean include (Ambrosino and Vitacca, Multidiscip Respir Med 2018, 13: 6; Davies et al., Br J Anaesth 2017, 118: 563-9):

- Increased work of breathing
- Impaired respiratory drive
- Inspiratory muscle weakness

67:

Long-Term Acute Care (LTAC) is a recognized level of care designation by the Centers for Medicare and Medicaid Services (CMS) for acute care hospitals. It is defined by federal statutes as a level of care for patients requiring an average length of stay of 25 days or more (Centers for and Medicaid Services, Fed Regist 2018, 83: 41144-784). Patients who require a short stay for treatment may be more appropriate for the subacute or skilled nursing facility (SNF) level of care.